

AADARSH MEHDI

Experienced Software Developer | Cloud-Native Platforms | AI/ML Systems

Budapest, Hungary | +36205176258 | aadarshofficial110@gmail.com

LinkedIn: <https://linkedin.com/in/aadarsh-mehdi-73754b13b> | GitHub: <https://github.com/amhkhawaja> | Website: <https://amhkhawaja.github.io>

PROFESSIONAL SUMMARY

Software Developer with 4+ years of experience designing, owning, and operating cloud-native microservices in telecom-scale production environments. End-to-end ownership of services from API design and implementation through testing, containerization, and production deployment on Kubernetes. Experienced in Python, Go, and Java backend ecosystems, secure coding practices, and database migration strategies. Additionally hands-on with GenAI, RAG pipelines, multi-agent AI systems and LLM Evaluation and Optimization Strategies. Eligible to work in Hungary without sponsorship.

TECHNICAL SKILLS

Programming Languages	Python, Go, Java, Kotlin, C++, JavaScript, TypeScript, Bash, PowerShell
Backend & Distributed Systems	Microservices, REST APIs, Event-Driven Architecture, Java Spring Boot, Kotlin Spring Boot, Apache Kafka, RabbitMQ, OpenAPI, Release Upgrades & Rollbacks, Schema/Data Migrations
Frontend	React, Streamlit, Gradio, HTML, CSS, Bootstrap
Web Frameworks	Flask, FastAPI, Spring Boot
Cloud & DevOps	AWS (EC2, EKS, IAM, S3, Lambda, CloudWatch, SQS), GCP (GKE), Terraform, Kubernetes, Docker, Helm, Jenkins, Apache Airflow, CI/CD
AI / ML / GenAI	LangChain, LangFlow, LangGraph, CrewAI, Instructor, LiteLLM, DeepEval, TensorFlow, Keras, RASA, NLP, RAG, Kiro CLI (AI coding agent), Multi-Agent AI Systems, Agentic AI Architecture, Multi-Agent Orchestration
Databases	PostgreSQL, MongoDB, Redis, InfluxDB, ETCD, OpenSearch
Observability & Security	Prometheus, Grafana, VictoriaMetrics, OAuth 2.0, RBAC, Keycloak, SCA, SonarQube, SecretOps
Testing & Automation	Robot Framework, Selenium
Version Control & Collaboration	Git, GitHub, GitLab, Gerrit, Jira, Confluence

PROFESSIONAL EXPERIENCE

Experienced Software Developer - Cloud LCM Microservices

Ericsson Hungary (Mar 2023 - Present)

- Designed, developed, and maintained 10+ scalable backend microservices using frameworks such as Python (FastAPI, Flask), Java (Spring Boot), and Go's native web framework.
- Owned 2 microservices end-to-end. Contributions included OpenAPI schema creation, API development, testing, maintenance, containerization, and deployment using Docker, Kubernetes, and Helm, ensuring higher availability, quality and timely delivery.
- Created shared libraries like auth, object_storage, search_engine and integrated it to all impacted microservices.
- Wrote unit tests, functional tests, integration tests, and end-to-end tests using pytest, JUnit, testify (Go), and Robot Framework. Reduced CI/CD test execution time by 40% through optimized test strategies.
- Engineered upgrade and rollback workflows for multi-service database migration from etcd to OpenSearch, implementing lifecycle hooks ensuring zero-data-loss rollback capability with backward-compatible deployment strategies
- Proactively conducted code reviews based on OWASP Top 10 and company internal secure coding guidelines. Integrated SCA (Software Composition Analysis) vulnerability checks into CI/CD pipelines.
- Architected and developed innovative GenAI projects for our product and team, leveraging LangChain, LangFlow, LangGraph, and Kiro agentic terminal.

Python Lab Instructor (Part-time)

Eötvös Loránd University (Sep 2023 - Jul 2025)

- Taught Python in Practice to 100+ students across 4 semesters.
- Covered topics like Python core, REST APIs using Flask and FastAPI, data analysis using data science libraries (NumPy, Pandas, Matplotlib), and machine learning fundamentals using scikit-learn and TensorFlow Keras.
- Supervised and mentored 5 final-semester student developers for their theses, guided them with industrial coding best practices, suggested feature implementation ideas, and resolved their queries about Generative AI concepts like RAG.

AI/ML Developer - IoT Accelerator Service Portal

Ericsson Hungary (Jul 2022 - Mar 2023)

- Independently architected and developed an ML + RASA-based Digital Assistant for the IoT service portal (an internal product).
- Performed data engineering for data collection, annotation, and augmentation using Python data science libraries. Created ETL

pipeline using Apache Airflow.

- Trained 5 different architecture models: 2 for NLU (intent classification and named entity recognition) using TensorFlow and SpaCy, and 3 for RASA core (stories, actions, and dialogue management).
- Developed custom action handlers in Python, invoked based on mappings of stories, NLU (intent classification and NER)
- Configured and used RabbitMQ service for message queues.
- Deployed the containerized application on Google Kubernetes Engine (GCP) and exhibited it at Ericsson Innovation Day to external customers.

Software Engineer Trainee - 3G Team

Nokia Hungary (Jan 2022 - Jul 2022)

- Refactored legacy build systems from Makefile to Meson to improve maintainability across 30+ repositories.
- Performed functional and integration testing using Robot Framework.
- Developed Bash scripts to perform root cause analysis on a complex product.

EDUCATION

B.Sc. in Computer Science - Eötvös Loránd University (ELTE), Budapest (2020-2023)

Thesis: "Adaptive, Context-Aware AI Conversational Agent for IoT Service Portal"

PROJECTS

Archloop.io (website live)

AI-powered SaaS platform for project management across engineering domains. Multi-tenant, multi-domain architecture that generates system designs, integrates with Jira/GitHub/GitLab, and provides domain-specific simulation and planning tools. Built with FastAPI, React, Keycloak, Nango, Temporal, Stripe, LangChain, and Haystack, Supabase. Features agentic AI orchestration with automatic failure recovery.

AI Auto Improver

Self-optimizing FastAPI microservice with automated prompt evolution, pairwise regression gates, multi-dimensional evaluation metrics, concurrent processing, provider-agnostic LLM orchestration, and containerized deployment. Tools: Python 3.12, FastAPI, Instructor, LiteLLM, Pydantic v2, SQLite, asyncio, Gradio, Docker, Mistral AI.

AWS ETL and ML pipeline Serverless

Serverless pipeline for preprocessing car sale data provisioned from terraform for AWS and Floci.io(AWS emulator) , used AWS services Lambda, S3, CloudWatch, SQS. and for ML used HistGradientBoostingRegressor model from sklearn.

Pay As You Go

SaaS-based microservice architecture project, developed in Kotlin Spring Boot, MongoDB, RabbitMQ, Keycloak for authentication, also with OAuth2.0 for social media accounts, RBAC authorization, Stripe for payments, and React for the frontend with three backend microservices.

Multi-Agent AI Configuration System

LangFlow-based agents generating and validating config changes between releases based on CPI documentation, NIR reports, and past release templates.

Autonomous AI Software Engineering Platform

Developed a pipeline-driven platform with tools and AI components to develop, test, containerize, and deploy applications to a Kubernetes cluster based on a prompt.

GenAI RAG Chatbot

Retrieval-augmented generation system for book-based question answering. Created RAG on the Alkafi book.

Intelligent IoT Digital Assistant

NLP-based conversational agent using TensorFlow, RASA, MongoDB.

ACHIEVEMENTS

- **Ericsson Innovation Day:** - Exhibited MVP Digital Assistant for IoT Accelerator with Ericsson Customers
- **Google Hashcode:** - 1st in ELTE, 20th in Hungary
- **CodeX Hackathon:** - Audience Award

WORK AUTHORIZATION

Hungarian Permanent Residence (Nemzeti Tartózkodási Kártya). Eligible to work without sponsorship.